

# Final Mock Exam — Solutions

## Lectures 1–12 (comprehensive)

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# Overview of the final mock exam

| Problem  | Topic                                                 | Points     |
|----------|-------------------------------------------------------|------------|
| P1       | Moving beyond linearity — splines (Ch. 7)             | 16         |
| P2       | Generalised additive models (Ch. 7)                   | 8          |
| P3       | Regression and classification trees by hand (Ch. 8)   | 20         |
| P4       | Ensembles — bagging, random forests, boosting (Ch. 8) | 16         |
| P5       | Neural networks by hand (Ch. 10)                      | 20         |
| P6       | Multiple testing (Ch. 13)                             | 20         |
| P7       | Cumulative essentials (Ch. 2–6)                       | 20         |
| $\Sigma$ |                                                       | <b>120</b> |

## Exam conditions

120 minutes for 120 points ( $\approx 1$  point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

## Problem 1 — task

### Problem 1 (16 P) — Moving beyond linearity

- (a) [4 P] Degrees of freedom of a cubic spline with  $K$  interior knots — justify by counting parameters and constraints; evaluate for  $K = 4$ .
- (b) [4 P] Degrees of freedom of a *natural* cubic spline with  $K$  knots ( $K = 4$ ); which extra constraint acts where and why is it useful?
- (c) [4 P] Smoothing spline  $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$ : role of the two terms; behaviour of  $\hat{g}$  as  $\lambda \rightarrow 0$  and  $\lambda \rightarrow \infty$ .
- (d) [4 P] Truncated-power basis of a cubic spline with one knot  $\xi$ ; define  $(x - \xi)_+^3$ ; why does adding it keep the fit smooth at  $\xi$ ?

# Problem 1 — solution (a) and (b)

## Solution P1 — (a) and (b)

**(a)**  $K$  knots  $\Rightarrow K + 1$  cubic pieces  $\times 4$  coefficients  $= 4K + 4$  parameters; continuity of  $g$  and  $g'$  and  $g''$  at each knot  $\Rightarrow 3K$  constraints:

$$\text{df} = 4(K + 1) - 3K = K + 4 \quad \Rightarrow \quad K = 4 : \text{df} = 8.$$

(Equivalently: basis  $1, x, x^2, x^3$  plus one truncated power per knot.)

**(b)** Natural spline: *linear beyond the boundary knots* (regions  $X < \xi_1$  and  $X > \xi_K$ ); killing the quadratic and cubic terms there costs 2 parameters per boundary region:

$$\text{df} = (K + 4) - 4 = K \quad \Rightarrow \quad K = 4 : \text{df} = 4.$$

*Why useful:* splines are most erratic near the boundaries where data are sparse; the linearity constraint stabilises the fit there (lower variance for a little extra bias).

# Problem 1 — solution (c) and (d)

## Solution P1 — (c) and (d)

(c) RSS term = fidelity to the data;  $\lambda \int g''(t)^2 dt$  penalises curvature (wiggleness);  $\lambda$  trades the two off.

- $\lambda \rightarrow 0$ : no penalty  $\Rightarrow \hat{g}$  *interpolates* the training points exactly — extremely wiggly, zero training RSS, severe overfitting (effective df  $\rightarrow n$ ).
- $\lambda \rightarrow \infty$ : curvature infinitely expensive  $\Rightarrow g'' \equiv 0 \Rightarrow \hat{g}$  is the *least-squares straight line* (effective df  $\rightarrow 2$ ).

$\lambda$  (chosen by LOOCV) steers the effective df smoothly between  $n$  and 2.

(d)  $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$  with

$$(x - \xi)_+^3 = \begin{cases} (x - \xi)^3 & x > \xi \\ 0 & x \leq \xi \end{cases}$$

$(x - \xi)_+^3$  has value and first and second derivative 0 at  $\xi \Rightarrow$  only the third derivative jumps: the fit stays continuous with continuous  $g'$  and  $g''$  — a cubic spline.

## Problem 2 — task

### Problem 2 (8 P) — Generalised additive models

- (a) [3 P] General form of a GAM for quantitative  $Y$ ; how does it extend multiple linear regression?
- (b) [3 P] One advantage concerning the interpretation of the  $f_j$ ; the main structural limitation — and its repair.
- (c) [2 P] Backfitting in one or two sentences.

## Problem 2 — solution

### Solution P2 — GAMs

**(a)**  $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p) + \varepsilon$ .

Linear regression is the special case  $f_j(X_j) = \beta_j X_j$ ; a GAM replaces each linear term by a smooth non-linear  $f_j$  (spline, local regression) but keeps the *additive* structure.

**(b)** *Advantage*: additivity  $\Rightarrow$  the effect of each  $X_j$  is its own function  $f_j$ , which can be plotted and interpreted individually while the other predictors are held fixed — one panel per predictor, as readable as a coefficient table.

*Limitation*: no *interactions* — the effect of  $X_j$  cannot depend on  $X_k$ .

*Repair*: add interaction components manually, e.g.  $X_j \times X_k$  or a two-dimensional smooth  $f_{jk}(X_j, X_k)$ .

**(c)** *Backfitting*: cycle through  $j = 1, \dots, p$  and re-fit each  $f_j$  to the partial residuals  $r_i = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik})$  with all other functions held fixed; repeat until convergence.

## Problem 3 — task

### Problem 3 (20 P) — Trees by hand

Advertising budget  $x$  (1,000 EUR) and weekly sales  $y$  (1,000 units) for  $n = 6$  products; one split at  $s$ :  $R_1 = \{x < s\}$  and  $R_2 = \{x \geq s\}$  with region-mean predictions.

| $i$   | 1 | 2 | 3 | 4 | 5 | 6  |
|-------|---|---|---|---|---|----|
| $x_i$ | 1 | 2 | 3 | 4 | 5 | 6  |
| $y_i$ | 2 | 4 | 3 | 3 | 8 | 10 |

(a) **[8 P]** Evaluate the cutpoints  $s = 2.5$  and  $s = 4.5$  by total RSS; which is chosen? (b) **[2 P]** Prediction for a new product with  $x = 5$ . (c) **[6 P]** Node with 6 “Yes” / 2 “No”: proportions; Gini; entropy ( $\ln 0.75 = -0.2877$ ;  $\ln 0.25 = -1.3863$ ); predicted class. (d) **[4 P]** Cost-complexity pruning: role of  $\alpha$  in  $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$ .



## Problem 3 — solution (a) and (b)

### Solution P3 — (a) and (b)

(a) Region means and RSS per cutpoint:

| $s$ | $R_1$ (y-values) | $\hat{y}_{R_1}$ | $R_2$ (y-values) | $\hat{y}_{R_2}$ | total RSS     |
|-----|------------------|-----------------|------------------|-----------------|---------------|
| 2.5 | {2,4}            | 3               | {3,3,8,10}       | 6               | $2 + 38 = 40$ |
| 4.5 | {2,4,3,3}        | 3               | {8,10}           | 9               | $2 + 2 = 4$   |

Details:  $s = 2.5$ :  $(2 - 3)^2 + (4 - 3)^2 = 2$ ;  $(3 - 6)^2 + (3 - 6)^2 + (8 - 6)^2 + (10 - 6)^2 = 9 + 9 + 4 + 16 = 38$ .

$s = 4.5$ :  $1 + 1 + 0 + 0 = 2$ ;  $(8 - 9)^2 + (10 - 9)^2 = 2$ .

Greedy recursive binary splitting picks the smaller RSS:  $s = 4.5$  ( $4 \ll 40$ ; root RSS was 52) — it isolates the two high-sales products.

(b)  $x = 5 \geq 4.5 \Rightarrow$  region  $R_2 \Rightarrow \hat{y} = \hat{y}_{R_2} = 9$  (i.e. 9,000 units per week).

## Problem 3 — solution (c) and (d)

### Solution P3 — (c) and (d)

(c)  $\hat{p}_{\text{Yes}} = 6/8 = 0.75$ ;  $\hat{p}_{\text{No}} = 2/8 = 0.25$ .

$$G = 0.75 \cdot 0.25 + 0.25 \cdot 0.75 = 0.375$$

$$D = -(0.75 \ln 0.75 + 0.25 \ln 0.25) = 0.75 \cdot 0.2877 + 0.25 \cdot 1.3863 = 0.2158 + 0.3466 = 0.562$$

Predicted class: majority = **Yes** (estimated probability 0.75). Both measures are near 0 for pure nodes — this node is moderately impure.

(d)  $\alpha$  = price per terminal node  $|T|$ :

- $\alpha = 0$ : criterion = training RSS  $\Rightarrow$  full tree  $T_0$  is optimal.
- $\alpha \uparrow$ : leaves must pay for themselves; branches with small RSS gain are pruned  $\Rightarrow$  nested sequence of ever smaller subtrees (weakest-link pruning).
- Choice: *cross-validation* over  $\alpha$ ; refit the chosen subtree on all data — balances bias against variance.

## Problem 4 — task

### Problem 4 (16 P) — Ensembles

- (a) [6 P]  $B$  identically distributed trees with variance  $\sigma^2$  and pairwise correlation  $\rho$ :  
 $\text{Var}\left(\frac{1}{B} \sum_b \hat{f}_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ . (i) Why does this show variance reduction and what limits it? (ii) Evaluate for  $B = 100$  and  $\rho = 0.6$  and  $\sigma^2 = 4$ ; compare with one tree. (iii) Limit as  $B \rightarrow \infty$ ?
- (b) [4 P] Random forests: modification at each split; recommended  $m$  for  $p = 25$ ; why does it improve on bagging (use the formula)?
- (c) [4 P] Boosting: the three tuning parameters; the slow-learning idea.
- (d) [2 P] What does the OOB error estimate and why is it free?

## Problem 4 — solution (a) and (b)

### Solution P4 — (a) and (b)

(a) (i) Averaging drives the second term  $\frac{1-\rho}{B}\sigma^2 \rightarrow 0$  (the independent part of the trees' variability); the first term  $\rho\sigma^2$  is independent of  $B$  — correlation between bootstrap trees limits the reduction.

(ii)  $\text{Var} = 0.6 \cdot 4 + \frac{0.4}{100} \cdot 4 = 2.4 + 0.016 = 2.416$  vs.  $\sigma^2 = 4$  for a single tree ( $\approx 40\%$  less).

(iii)  $B \rightarrow \infty$ : variance  $\rightarrow \rho\sigma^2 = 2.4$  (floor; more trees never overfit, they only stabilise).

(b) At *each split* only a random subset of  $m$  of the  $p$  predictors may be used; classification default  $m \approx \sqrt{p}$   
 $\Rightarrow p = 25$ :  $m = \sqrt{25} = 5$ .

With bagging a dominant predictor tops nearly every tree  $\Rightarrow$  trees alike  $\Rightarrow \rho$  high. Restricting candidates *decorrelates* the trees ( $\rho \downarrow$ ), which lowers the floor  $\rho\sigma^2$  — the part of the variance that averaging alone cannot remove.

## Problem 4 — solution (c) and (d)

### Solution P4 — (c) and (d)

(c) Boosting for regression trees:

- $B$  — **number of trees**: added sequentially; too many  $\Rightarrow$  slow overfitting; choose by CV.
- $\lambda$  — **shrinkage** (e.g. 0.01 or 0.001): scales each new tree's contribution; smaller  $\lambda \Rightarrow$  slower learning  $\Rightarrow$  larger  $B$  needed.
- $d$  — **splits per tree** (interaction depth): complexity and interaction order of each tree; often stumps ( $d = 1$ ) suffice (additive model).

*Slow learning*: each tree is fitted to the current *residuals* and only the fraction  $\lambda$  is added — the model improves gradually exactly where it still errs, instead of fitting the data hard at once; this tends to generalise better.

(d) Each bootstrap sample misses  $\approx 1/3$  of the observations (out-of-bag); predicting every observation only with trees that never saw it yields the **OOB error** — a valid *test-error* estimate, free as a by-product of the bootstrap (no validation set or CV loop needed).

## Problem 5 — task

### Problem 5 (20 P) — Neural networks by hand

Input  $x = (1, 2)$ ; hidden ReLU units  $g(z) = \max\{0, z\}$ ; linear output:

$$A_1 = g(-1 + 1x_1 + 2x_2) \quad A_2 = g(-3 - 2x_1 + 1x_2) \quad f(x) = 1 + 2A_1 + 4A_2$$

- (a) **[8 P]** Forward pass step by step: pre-activations;  $A_1$  and  $A_2$ ; output  $f(x)$ . What did the ReLU do to unit 2 and why are non-linear activations essential?
- (b) **[6 P]** Logits  $z = (1.0, 0.0, -1.0)$ ; true class 1: all three softmax probabilities and the cross-entropy loss ( $e^1 = 2.718$ ;  $e^0 = 1.000$ ;  $e^{-1} = 0.368$ ;  $\ln 4.086 = 1.408$ ).
- (c) **[4 P]** Parameter count of a  $p = 4 \rightarrow k = 3 \rightarrow K = 2$  network with biases (show both layers).
- (d) **[2 P]** Conv layer:  $32 \times 32$  input;  $F = 5$ ;  $P = 2$ ;  $S = 1$ : output size via  $(W - F + 2P)/S + 1$ .

## Problem 5 — solution (a) forward pass

### Solution P5 — (a) forward pass

Pre-activations:

$$z_1 = -1 + 1 \cdot 1 + 2 \cdot 2 = 4$$

$$z_2 = -3 - 2 \cdot 1 + 1 \cdot 2 = -3$$

ReLU activations:

$$A_1 = \max\{0, 4\} = 4$$

$$A_2 = \max\{0, -3\} = 0$$

Output:

$$f(x) = 1 + 2 \cdot 4 + 4 \cdot 0 = 9$$

The ReLU *switched unit 2 off*: its negative pre-activation is clipped to 0, so it contributes nothing for this input (though it can for others).

*Why essential*: with a linear  $g$  the whole network collapses into one linear function of  $x$  regardless of depth — the non-linearity is what lets hidden layers build flexible functions (piecewise-linear surfaces with ReLU).

## Problem 5 — solution (b) softmax and cross-entropy

### Solution P5 — (b) softmax and cross-entropy

Exponentials and their sum:

$$e^{1.0} = 2.718 \quad e^{0.0} = 1.000 \quad e^{-1.0} = 0.368 \quad \Rightarrow \quad 2.718 + 1.000 + 0.368 = 4.086$$

Softmax probabilities:

$$\hat{p}_1 = \frac{2.718}{4.086} = 0.665 \quad \hat{p}_2 = \frac{1.000}{4.086} = 0.245 \quad \hat{p}_3 = \frac{0.368}{4.086} = 0.090$$

(sum = 1.000 ✓). Cross-entropy for true class 1:

$$-\ln \hat{p}_1 = -\ln \frac{e^1}{4.086} = -(1 - \ln 4.086) = 1.408 - 1 = 0.408$$

Sanity checks: a perfect prediction ( $\hat{p}_1 = 1$ ) gives loss 0; predicting the uniform  $1/3$  would give  $\ln 3 \approx 1.099$  — here the network is right and fairly confident.



## Problem 5 — solution (c) and (d)

### Solution P5 — (c) parameters and (d) convolution

(c) Fully connected  $4 \rightarrow 3 \rightarrow 2$  with biases:

| Layer                       | Weights          | Biases | Parameters |
|-----------------------------|------------------|--------|------------|
| input $\rightarrow$ hidden  | $4 \cdot 3 = 12$ | 3      | 15         |
| hidden $\rightarrow$ output | $3 \cdot 2 = 6$  | 2      | 8          |
| total                       |                  |        | 23         |

(d) Output width =  $\frac{W - F + 2P}{S} + 1 = \frac{32 - 5 + 2 \cdot 2}{1} + 1 = 31 + 1 = 32 \Rightarrow$  output feature map  $32 \times 32$ :  
with  $F = 5$  and  $P = 2$  and  $S = 1$  this is the “same” convolution — spatial size preserved (one map per filter).

## Problem 6 — task

### Problem 6 (20 P) — Multiple testing

- (a) **[4 P]**  $m = 20$  independent tests at  $\alpha = 0.05$ ; all nulls true: define and compute the FWER  $((0.95)^{20} = 0.3585)$ ; interpret.
- (b) **[7 P]** Six ordered  $p$ -values 0.001; 0.004; 0.011; 0.030; 0.045; 0.230 — FWER control at  $\alpha = 0.05$  with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?
- (c) **[6 P]** Benjamini–Hochberg at  $q = 0.05$  on the same  $p$ -values: thresholds  $jq/m$ ; rejections; what exactly does FDR control guarantee?
- (d) **[3 P]** Exploratory screen of 20,000 genes with follow-up validation: FWER or FDR? Justify.

## Problem 6 — solution (a) FWER

### Solution P6 — (a) FWER for 20 independent tests

FWER =  $\Pr(V \geq 1)$  — the probability of *at least one* false rejection among the  $m$  tests ( $V$  = number of falsely rejected true nulls).

With all 20 nulls true and independent tests at level 0.05:

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{20} = 1 - (0.95)^{20} = 1 - 0.3585 = 0.6415$$

*Interpretation:* although each test individually errs with probability 5% only, the chance of at least one spurious “discovery” among 20 unadjusted tests is about 64% — multiple testing without adjustment almost guarantees false findings (with  $m = 100$ :  $1 - 0.95^{100} \approx 0.994$ ).

## Problem 6 — solution (b) Bonferroni and Holm

### Solution P6 — (b) Bonferroni and Holm at level 0.05

**Bonferroni:** reject iff  $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$ : only 0.001 and 0.004 pass ( $0.011 > 0.00833$ )  $\Rightarrow$  reject  $H_{(1)}$  and  $H_{(2)}$ : **2 rejections**.

**Holm (step-down):** compare  $p_{(j)}$  with  $\alpha/(m - j + 1)$ ; stop at first failure:

| $j$                  | 1       | 2      | 3      | 4           | 5      | 6      |
|----------------------|---------|--------|--------|-------------|--------|--------|
| $p_{(j)}$            | 0.001   | 0.004  | 0.011  | 0.030       | 0.045  | 0.230  |
| $\alpha/(m - j + 1)$ | 0.00833 | 0.0100 | 0.0125 | 0.0167      | 0.0250 | 0.0500 |
| decision             | ✓       | ✓      | ✓      | <b>stop</b> | —      | —      |

$0.011 \leq 0.0125$  but  $0.030 > 0.0167 \Rightarrow$  reject  $H_{(1)}$  and  $H_{(2)}$  and  $H_{(3)}$ : **3 rejections**.

**Holm  $\supseteq$  Bonferroni:** Holm's hurdles  $\alpha/(m - j + 1) \geq \alpha/m$  for all  $j$  (equality only at  $j = 1$ )  $\Rightarrow$  uniformly more powerful — and it still controls the FWER without independence assumptions.

## Problem 6 — solution (c) and (d)

### Solution P6 — (c) Benjamini–Hochberg and (d) FDR vs FWER

(c) Thresholds  $jq/m = j \cdot 0.05/6$ :

| $j$                  | 1       | 2      | 3      | 4          | 5      | 6      |
|----------------------|---------|--------|--------|------------|--------|--------|
| $p_{(j)}$            | 0.001   | 0.004  | 0.011  | 0.030      | 0.045  | 0.230  |
| $jq/m$               | 0.00833 | 0.0167 | 0.0250 | 0.0333     | 0.0417 | 0.0500 |
| $p_{(j)} \leq jq/m?$ | yes     | yes    | yes    | <b>yes</b> | no     | no     |

Largest such  $j$ :  $L = 4$  ( $0.030 \leq 0.0333$ ; note  $0.045 > 0.0417$ )  $\Rightarrow$  reject *all* of  $H_{(1)}$  to  $H_{(4)}$ : **4 rejections**.  
*Guarantee*:  $\text{FDR} = E[V/\max(R,1)] \leq q$  — *on average* at most 5% of rejected hypotheses are false discoveries (no bound on  $\Pr(V \geq 1)$ ; that is FWER).

(d) Control the **FDR**. Bonferroni at  $0.05/20,000 = 2.5 \cdot 10^{-6}$  would leave almost no power — guarding against a *single* false positive is needlessly strict when every hit is re-tested anyway. FDR control tolerates a small known fraction of false leads in exchange for a much richer list of candidates — exactly right for exploratory screening with follow-up validation.

## Problem 7 — task

### Problem 7 (20 P) — Cumulative essentials

- (a) [4 P] True or false: “more flexible always means lower *test* error”. Justify.
- (b) [4 P] Logistic regression for default:  $\hat{\beta}_1 = 0.7$  for missed payments ( $e^{0.7} = 2.014$ ): effect of one more missed payment on the *odds*; why is there no single number for the effect on the *probability*?
- (c) [4 P] One reason to prefer 5- or 10-fold CV over LOOCV and one over a single validation split.
- (d) [2 P]  $p = 2,000$  predictors; only a handful truly matter: ridge or lasso? Why?
- (e) [6 P] Match to {linear regression; lasso; logistic regression; random forest} — each exactly once: (i) price per m<sup>2</sup> with CI; (ii) drug response from 5,000 genes ( $n = 100$ ) with a short gene list; (iii) default yes/no with odds reported to the regulator; (iv) machine failures from hundreds of interacting sensors; accuracy only.

## Problem 7 — solution (a) to (c)

### Solution P7 — (a) to (c)

**(a) False.** Test error =  $\text{bias}^2 + \text{variance} + \text{irreducible error}$ : flexibility lowers bias but raises variance; past the sweet spot the test error *rises* (U-shape). E.g. near-linear truth or small  $n$ : linear regression beats a flexible overfitter. (Only *training* error always falls with flexibility.)

**(b)** One more missed payment adds 0.7 to the log-odds, i.e. *multiplies* the odds of default by  $e^{0.7} = 2.014$  (roughly doubles them) — the same factor at every  $x$ . The *probability* change depends on the position on the S-curve: large near  $p \approx 0.5$ , tiny near 0 or 1  $\Rightarrow$  no single number.

**(c)** *vs. LOOCV*: the  $n$  leave-one-out fits are nearly identical  $\Rightarrow$  their errors are highly correlated  $\Rightarrow$  the LOOCV average has *higher variance* (and costs  $n$  fits). *vs. validation split*: a single split is highly variable and overestimates test error since only part of the data trains the model (*bias*);  $k$ -fold trains on  $(k-1)/k$  of the data and averages  $k$  estimates  $\Rightarrow$  less bias and more stability.

## Problem 7 — solution (d) and (e)

### Solution P7 — (d) and (e)

(d) **Lasso**: the  $\ell_1$  penalty sets coefficients *exactly to zero*  $\Rightarrow$  variable selection; ideal for a sparse truth (a handful of signals among 2,000) and yields a small interpretable model. Ridge keeps all 2,000 predictors.

(e) Matching (each method once):

- (i) **Linear regression** — quantitative response; goal is *inference*: effect size with confidence interval from OLS standard errors.
- (ii) **Lasso** —  $p = 5,000 \gg n = 100$ : OLS infeasible;  $\ell_1$  selects a short list of genes.
- (iii) **Logistic regression** — binary response; coefficients give odds ratios  $e^{\hat{\beta}_j}$ : interpretable for the regulator.
- (iv) **Random forest** — automatic non-linearities and interactions; strong accuracy with many mixed predictors; interpretability not needed.